

Mid-term Examination
– Introduction to Data Analytics –

Instructor: Pr. Youness Moukafih

Exam Duration: 2 hours

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Exercise 1. (1.5 mark).

Let X be a continuous, strictly positive variable ($x_i > 0$). Define

$$S = \frac{\text{median}(X) - Q_1}{Q_3 - \text{median}(X)},$$

where Q_1 and Q_3 are the first and third quartiles of X . Use S as a skewness heuristic. What is the expected shape of the distribution in each case? Explain your reasoning.

(a) $S = 1$

(b) $S < 1$

(c) $S > 1$

Exercise 2. (2 marks).

Ice cream usually comes in 1.5 quart boxes (48 fluid ounces), and ice cream scoops hold about 2 ounces. However, there is some variability in the amount of ice cream in a box as well as the amount of ice cream scooped out. We represent the amount of ice cream in the box as X and the amount scooped out as Y . Suppose these random variables have the following means, standard deviations, and variances:

variable	mean (oz)	SD	variance
X	48	1	1
Y	2	0.25	0.0625

- An entire box of ice cream, plus 3 scoops from a second box is served at a party. How much ice cream do you expect to have been served at this party? What is the standard deviation of the amount of ice cream served?
- How much ice cream would you expect to be left in the box after scooping out one scoop of ice cream? That is, find the expected value of $X - Y$. What is the standard deviation of the amount left in the box?
- Using the context of this exercise, explain why we *add variances* when we subtract one random variable from another.
- Suppose we have n independent observations $X_1, X_2, \dots, X_n$ from a distribution with mean μ and standard deviation σ . What is the variance of the mean of these n values,

$$\frac{X_1 + X_2 + \dots + X_n}{n}?$$

Exercise 3. (8 marks).

Consider the following table with two variables: shape and color:

Shape	Red	Blue	Green	Yellow
Circle (C)	0.07	0.09	0.08	0.06
Triangle (T)	0.11	0.13	0.05	0.06
Square (Sq)	0.06	0.12	0.10	0.07

In this exercise we mean by **XOR**: “exactly one of A and B occurs (A or B, but not both).”

Named events:

$$E = \{\text{edge-bearing shape}\} = \{T \text{ or Sq}\}, \quad U = \{\text{high-symmetry shape}\} = \{C \text{ or Sq}\},$$

$$B = \{\text{blue color}\} = \{\text{Blue}\}, \quad D = \{\text{cool color}\} = \{\text{Blue or Green}\}.$$

Bayes’ theorem (for use in part (g)):

$$\Pr(H_i | D) = \frac{\underbrace{\Pr(D | H_i)}_{\text{likelihood}} \underbrace{\Pr(H_i)}_{\text{prior}}}{\underbrace{\sum_j \Pr(D | H_j) \Pr(H_j)}_{\text{evidence (marginal likelihood)}}} \Rightarrow \underbrace{\Pr(H_i | D)}_{\text{posterior}}.$$

- (a) Compute $\Pr(E)$, $\Pr(B)$, $\Pr(D)$, and $\Pr(U)$ directly from the table.
 - (b) Compute the probability that *exactly one* of E and B occurs (XOR) in two different ways:
 - (a) By summing all cells that satisfy (E and not B) or (B and not E).
 - (b) By first finding $\Pr(E \text{ or } B) = \Pr(E) + \Pr(B) - \Pr(E \text{ and } B)$, then converting it to the XOR probability.
- Verify the two results match.
- (c) Compare directions of conditioning by computing $\Pr(E | B)$ and $\Pr(B | E)$. Report their ratio $\Pr(E | B) / \Pr(B | E)$ and give a one-sentence interpretation.
 - (d) Using only $\Pr(E)$ and $\Pr(B)$, derive the Fréchet bounds for $\Pr(E \text{ and } B)$. From those bounds, deduce the tightest possible bounds for the probability that *exactly one* of E and B occurs. Compare these bounds to the actual values computed from the table.
 - (e) Test conditional independence within a structural slice: are E and B independent given U? Compute and compare $\Pr(E \text{ and } B | U)$ with $\Pr(E | U) \Pr(B | U)$. State which is larger (or if equal) and interpret briefly.
 - (f) Two independent draws are made. What is the probability the two draws have the *same color* but *different shapes*? First express your answer as a sum over colors, then evaluate numerically.
 - (g) Using the Bayes formula above, compute the posterior odds between Triangle and Square given Blue:

$$\frac{\Pr(T | B)}{\Pr(\text{Sq} | B)}.$$

Then re-express the same quantity as

$$\frac{\Pr(B | T)}{\Pr(B | Sq)} \times \frac{\Pr(T)}{\Pr(Sq)}$$

and verify numerically that both approaches agree. Provide a one-sentence interpretation.

- (h) Compute the probability that *exactly one* of the three events E, B, U occurs.

Exercise 4. (6 marks).

In each round you roll two *fair* dice. If the two faces differ, you **lose \$1**. If they are the same (a *double*), you **win \$N**. Rounds are independent.

- (a) Find the value of N that makes a *single round* fair (expected net payoff = 0). With this N , compute the *variance* and *standard deviation* of the one-round payoff.
- (b) Play m independent rounds with the fair value of N from (a) and no other modifications. Let W_m be the total net winnings after m rounds.
 - (a) Express W_m in terms of the number of doubles D_m observed in m rounds.
 - (b) Derive a closed-form expression (as a binomial tail) for $\Pr(W_m > 0)$.
 - (c) Give conditions on m under which $\Pr(W_m > 0) \geq \frac{1}{2}$, and justify your reasoning.
- (c) Modify the round as follows: on any double other than (6,6), you still win N ; on (6,6) you win $N + J$ (a *jackpot* bonus $J > 0$). In addition, the house charges a fixed *per-round fee* c (paid every round, regardless of outcome).
 - (a) With the fair N from (a), find c (in terms of J) that makes a round *exactly fair*.
 - (b) Under that fair c , compute the *variance* of the one-round payoff as a function of J and determine whether it is increasing or decreasing in J .
- (d) Using the fair N from (a) and no fee ($c = 0$), you now *stop* at the first double or after K rounds, whichever comes first. Let $W^{(K)}$ be the total net winnings under this stopping rule.
 - (a) Show (either by direct summation or by an optional-stopping style argument) that $\mathbb{E}[W^{(K)}] = 0$ for every finite K .
 - (b) Express $\Pr(W^{(K)} = -K)$ and $\Pr(W^{(K)} = N - (j - 1))$ for $j = 1, \dots, K$ in closed form.

Exercise 5. (2.5 marks).

In this exercise we use the KL formula to quantify the distance between two discrete distributions:

$$D_{\text{KL}}(P \| Q) = \sum_i P(i) \log_2 \frac{P(i)}{Q(i)}.$$

Outcomes: $\{A, B, C\}$ **Distributions:**

$$P = (0.50, 0.25, 0.25), \quad Q = (0.25, 0.50, 0.25).$$

Q1. Compute $D_{\text{KL}}(P\|Q)$. Write the three terms for A, B, C separately, then sum.

Q2. Compute $D_{\text{KL}}(Q\|P)$ and compare to Q1. Is it the same number *for this example*? (Yes/No) Is KL divergence *always* symmetric? (Yes/No)

Q3. Zero contribution & zero divergence.

- Which outcome(s) contribute 0 to the sum, and why?
- In one sentence: when does $D_{\text{KL}}(P\|Q) = 0$?